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- module9_lab2.ipynb
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Migration: Moving from Google Colab to AWS SageMaker Studio Lab

These are the required steps, reusable actions, and risks associated with moving our lab's data science workflow from Google Colab to AWS SageMaker Studio Lab. Which is free, and cloud based.

1. What Transfers Are Untouched during a Migration?

Both platforms are built as open-source Jupyter Notebooks. Each project will migrate without any code modifications. The .ipynb files and notebook structure, such as the text cells, markdown formatting, and cell execution will transfer exactly. The Python Code such as, control loops, and data processing are the same. The Libraries that work for python code like pandas, numpy, scikit-learn, and matplotlib will run the same.

2. What Needs Redoing

The Cloud notebook environments handles authentication, file systems, and copies and downloads differently.

Regarding Accounts and Authentication, and the Platform Access, you must register for an AWS SageMaker Studio Lab account using your specific, school permission, working email. This operates independently of standard AWS cloud credentials. The API Credentials, and any access keys use SageMaker's terminal.

Regarding the File Storage And Data Access, you remove Google code uploading a dataset, on SageMaker. The Storage for SageMaker Studio Lab provides a permanent 15 GB block storage across sessions. You must upload files directly into this directory using the JupyterLab file interface or fetch data with wget or curl.

Regarding Packages Installment, unlike Colab, where !pip install must be rerun every time the notebook disconnects, SageMaker Studio Lab saves your environment. Using a Conda Environment, instead of running inline pip commands in your notebook, you will use the integrated Jupyter bash terminal to build an environment. Here is an example code snippet found on the SageMaker website.

conda create --name lab_env python=3.10 conda activate lab_env pip install -r requirements.txt

3. The Risks of a Migration, which is most critical is the temporary loss of data access limits and potential compute downtime. Unlike Google Colab's easily upgradeable runtime, SageMaker Studio Lab enforces a strict 15 GB storage max and restricts compute sessions to 8 hours for CPU or 4 hours for GPU per day. If your dataset exceeds 15 GB, or if your model training sequences require long, runtimes, the notebook will halt, resulting in lost training checkpoints.

4. There are more details to know about AWS. In Amazon Web Services there is an on demand cloud named the EC2 , which is accessed through the Management Console, or command line tools. The first thing needed is an account including phone number, address, and credit card information. The service has an initial free tier with imits. The EC2 instance must be launched to create a virtual machine. The options include choosing the AMI, a preferred OS, an instance type lke t3.micro, security group, creation of a key pair, for logging in, and the launch. Besides logging into your instance SSH using Putty, AWS provides an IDE called Cloud9. This is under developers tools. Create a Cloud9 environment, with a name, select "no-ingress" environment and t2.micro and Amazon Linux 2 platform. The bash terminal shell is like a Linux machine.
5. Unlike Google Coloab, a Python installation does not come with libraries. You need to install these, in the bash terminal. "pip install ..." You also have to access training data for machine learning models. Use the command line tools for this. "wget then the URL to the desired data" for a download to your current directory.
6. Graphical output requires the file to be saved first. "plt.savefig(...)" not with plt.show like Google colabs. To view this select the file from the browser on the left of the dashboard.
7. Colab is not always a preferred way to do a data science lab, because easy access is available on the terminal of AWS Cloud9.
8. Cloud9 is only free to a certain staorage and processing limit, but Google Colab is always free, but can run out of RAM.
9. Moving between cloud platforms is a migration, when you use Google Colabs it's completely free (Colaboratory) Code is run on resources that are owned by Google, all you need is an active google account. Navigate to Google Drive, if you have Colab installed select "New", "More" Google Colabs will appear or install it as an App. The "New" button on colabs creates a new notebook. You can start with Python3. The python cells can be run individually, the trash symbol deletes unwanted code snippets you've written.
10. Many data science libraries come with colabs, like pandas, numpy, scikit-learn already installed.
11. A Word of Caution regarding a migration, involves leaving your files unmoved to your own machine and left on the cloud too long. a. On google Colabs data is automatically moved to trash after a time. b. On AWS data if data is taking up storage and you are not careful it can become costly. Vampire charges can occur if resources are not deleted properly, or when they exceed the free tier limits.